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Prenatal Exposure to Acid-Suppressive Medications and Risk of Neuropsychiatric Disorders in Children

Seohyun Hong; Sooji Lee, MD; Hyunjee Kim, MS; Hyesu Jo, MS; Kyeongmin Lee, MS; Yeona Jo, MS; Tae Hyeon Kim, MD; Jaeyu Park, MS; Jinseok Lee, PhD; Ho Geol Woo, MD, PhD; Hayeon Lee, PhD; Dong Keon Yon, MD, PhD

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IMPORTANCE Although acid-suppressive medications are commonly prescribed during pregnancy, comprehensive studies on their association with neuropsychiatric disorders in children are limited.

OBJECTIVE To investigate potential associations between prenatal exposure to acid-suppressive medication (histamine 2 [H_2] receptor antagonists and proton pump inhibitors [PPIs]) and neuropsychiatric disorders in children.

DESIGN, SETTING, AND PARTICIPANTS Retrospective cohort study of mother-child pairs in the South Korean National Health Insurance Service database with births from January 2010 through December 2017, with follow-up of offspring through December 2023. Analyses were conducted using a propensity score–based overlap-weighted cohort and a sibling-matched cohort.

EXPOSURE Prenatal exposure to at least 1 prescription for a PPI or H_2 receptor antagonist.

MAIN OUTCOMES AND MEASURES Offspring neuropsychiatric disorders identified via *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* codes, including attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorders (ASD), intellectual disability, severe neuropsychiatric disorder, and obsessive-compulsive disorder.

RESULTS Of 3 012 992 mother-child pairs initially identified, 2 777 119 were included, with 507 845 pairs exposed to prenatal acid-suppressive medication and were followed up a mean of 10.3 (SD, 2.3) years. In the overlap-weighted cohort (403 658 mother-child pairs exposed and 403 659 pairs unexposed to an acid-suppressive medication), respective risks among offspring exposed vs not exposed were 4.85% vs 4.25% for ADHD, 1.45% vs 1.33% for ASD, 1.25% vs 1.09% for intellectual disability, 0.94% vs 0.81% for severe neuropsychiatric disorder, and 0.30% vs 0.27% for obsessive-compulsive disorder. In the overlap-weighted cohort, the adjusted hazard ratio (HR) following prenatal acid-suppressive medication exposure was 1.14 (95% CI, 1.12-1.17) for ADHD, 1.07 (95% CI, 1.03-1.11) for ASD, 1.13 (95% CI, 1.09-1.18) for intellectual disability, 1.16 (95% CI, 1.10-1.21) for severe neuropsychiatric disorder, and 1.12 (95% CI, 1.03-1.21) for obsessive-compulsive disorder. Sibling-control analyses that included 157 069 exposed and 164 669 unexposed offspring revealed no significant associations between prenatal exposure and offspring outcomes. The adjusted HR for ADHD was 0.98 (95% CI, 0.95-1.02); for ASD, 0.98 (95% CI, 0.92-1.04); intellectual disability, 1.02 (95% CI, 0.95-1.09); severe neuropsychiatric disorder, 1.00 (95% CI, 0.93-1.08); or obsessive-compulsive disorders, 0.95 (95% CI, 0.82-1.10).

CONCLUSIONS AND RELEVANCE Exposure to acid-suppressive medication during pregnancy was not associated with children's risk of ADHD, severe neuropsychiatric disorder, obsessive-compulsive disorder, intellectual disability, or ASD in sibling-control analyses. Small associations were observed in overlap-weighted models; these may reflect confounding by shared familial factors.

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Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Authors: Dong Keon Yon, MD, PhD, Department of Pediatrics, Kyung Hee University College of Medicine, 23 Kyungheedae-ro, Dongdaemun-gu, Seoul 02447, South Korea (yonkkang@gmail.com); Hayeon Lee, PhD, Department of Biomedical Engineering, Kyung Hee University, 1732 Deogyeong-daero, Giheung-gu, Yongin 17104, South Korea (wwhy28@khu.ac.kr).

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Acid-suppressive medications, including proton pump inhibitors (PPIs) and histamine 2 (H_2) receptor antagonists, are commonly prescribed during pregnancy to manage heartburn and gastroesophageal reflux disease.[1,2] Recent studies have raised concerns regarding associations between prenatal acid-suppressive medication exposure and adverse outcomes in infants or children, including increased risks of preterm birth,[3] allergic diseases,[4] asthma,[5] and major congenital malformations.[6] One study identified an association between prenatal PPI exposure and epilepsy in offspring, raising the question of whether acid-suppressive medications may affect the developing nervous system.[7]

The early-life gut microbiome has been described as a critical factor in neurodevelopmental health based on the premise that microbial colonization during fetal and neonatal periods plays a fundamental role in shaping brain circuit maturation. Previous studies have suggested that alteration in the gut microbiome can adversely impact neurodevelopment.[8,9] Considering that acid-suppressive medications may modify the composition and diversity of the gut microbiota,[10] maternal use during pregnancy may influence neurodevelopmental processes in offspring.[11] Furthermore, a 2022 study suggested that early administration of acid-suppressive medications before 2 years of age in preterm infants could be associated with increased risks of neurodevelopmental disorders.[12] This raises concern that similar mechanisms might occur during in utero exposure because PPIs and H_2 receptor antagonists readily cross the placental barrier and may thereby influence fetal neurodevelopment.

The potential association between prenatal acid-suppressive medications exposure and major neuropsychiatric disorders remains insufficiently investigated. As the use of acid-suppressive medications during pregnancy continues to increase,[13] assessment of their association with long-term neurodevelopment is warranted. This study aimed to investigate the association between maternal acid-suppressive medication use and the risk of neuropsychiatric disorders in children, using a nationwide, population-based, large-scale birth cohort in South Korea.

Key Points

Question Is prenatal acid-suppressive medication exposure associated with neuropsychiatric disorders in offspring?

Findings In this cohort study of more than 2.7 million mother-child pairs, models without sibling controls identified modestly increased risks of neuropsychiatric disorders in children associated with acid-suppressive medication use during pregnancy (adjusted hazard ratio [HR], 1.13). However, sibling-matched analyses did not identify evidence of increased risk of any disorder (adjusted HR, 1.0), including attention-deficit/hyperactivity disorder, autism spectrum disorders, intellectual disability, severe neuropsychiatric disorder, and obsessive-compulsive disorder.

Meaning Prenatal exposure to acid-suppressive medication was not associated with children's risk of neuropsychiatric disorders in sibling-control analyses; associations observed in other models may reflect unmeasured confounding.

Methods

Data Source

This study used the large-scale, nationwide mother-child linked cohort provided by the National Health Insurance Service (NHIS) of South Korea, from January 1, 2009, through December 31, 2023 (total mother-child pairs, 3 012 992). The NHIS operates a single-payer, mandatory universal health insurance system covering approximately 97% of the Korean population.[14] The NHIS database contains comprehensive information on sociodemographic characteristics, diagnoses, procedures, prescriptions, health examinations, and vital statistics for all insured individuals.[14]

Each mother-child linkage was established using a unique family insurance identification number. The start of pregnancy was estimated using a validated algorithm based on delivery dates and/or diagnoses related to preterm birth to calculate gestational age in administrative claims data. The dataset included deidentified information on patient demographics (eg, sex, age, region, insurance type, and income level), clinical diagnoses (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision [ICD-10]*, diagnosis date, medical specialty, and setting), medical procedures (surgical and nonpharmacological interventions), prescription records (drug codes and route of administration), and health care utilization (inpatient and outpatient visits). Additionally, data on infant health screening indicators (birth weight and preterm birth) were available.

This study was approved by the institutional review board of Kyung Hee University (KHSIRB-24-511[EA]) and the Korean NHIS (NHIS-2025-03-1-040). The requirement for informed consent was waived because only deidentified administrative data were used.[15] This study followed the Reporting of Studies Conducted Using Observational Routinely-Collected Health Data (RECORD) statement (eTable 1 in Supplement 1).

Study Population

The study population comprised women aged 13 to 58 years whose liveborn neonates were delivered from January 1, 2010, through December 31, 2017. The look-back period for assessing maternal characteristics was 12 months prior to the onset of pregnancy, and all offspring were followed up through December 31, 2023 (**Table 1**).

Participants with inadequate information on socioeconomic status and mothers with at least 1 prescription for an acid-suppressive medication within 30 days before pregnancy were excluded.

We constructed a propensity score-based overlap-weighted cohort to balance covariates and mitigate confounding bias between groups defined by acid-suppressive medication use (**Figure 1**).[16,17] In addition, we created a sibling-comparison cohort to assess whether observed associations changed after control for family-level confounders.[18]

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Table. Baseline Characteristics of Offspring in the Overlap-Weighted Cohort by Acid-Suppressive Medications Exposure

Characteristic	Unadjusted cohort (n = 2 777 119)[a]			Overlap-weighted cohort (n = 807 317)[a]		
	Exposed (n = 507 845)	Unexposed (n = 2 269 274)	SMD[b]	Exposed (n = 403 658)	Unexposed (n = 403 659)	SMD[b]
Covariates used in matching (mother)						
Age, y						
Mean (SD)	31.7 (4.3)	31.9 (4.1)	0.048	31.7 (3.8)	31.8 (1.8)	0.010
No. (%)						
<20	1869 (0.4)	5317 (0.2)	0.073	1324 (0.3)	1324 (0.3)	<0.001
20-24	26 210 (5.2)	95 748 (4.2)		19 952 (4.9)	19 952 (4.9)	
25-29	115 961 (22.8)	487 450 (21.5)		91 177 (22.6)	91 177 (22.6)	
30-34	237 792 (46.8)	1 105 323 (48.7)		190 806 (47.3)	190 806 (47.3)	
≥35	126 013 (24.8)	575 436 (25.4)		100 399 (24.9)	100 400 (24.9)	
Urban residence, No. (%)	230 690 (45.4)	1 003 013 (44.2)	0.025	182 403 (45.2)	182 404 (45.2)	<0.001
Household income, percentile, No. (%)						
Low (<25th)	132 591 (26.1)	548 680 (24.2)	0.049	103 498 (25.6)	103 499 (25.6)	<0.001
Lower middle (25th-49th)	120 704 (23.8)	539 993 (23.8)		96 042 (23.8)	96 042 (23.8)	
Upper middle (50th-74th)	146 919 (28.9)	678 873 (29.9)		117 736 (29.2)	117 736 (29.2)	
High (≥75th)	107 631 (21.2)	501 728 (22.1)		86 382 (21.4)	86 382 (21.4)	
Parity, No. (%)						
1	314 947 (62.0)	1 503 248 (66.2)	0.088	253 868 (62.9)	253 869 (62.9)	<0.001
≥2	192 898 (38.0)	766 026 (33.8)		149 790 (37.1)	149 790 (37.1)	
Prepregnancy maternal medical conditions, No. (%)						
Gastrointestinal disorder	173 947 (34.3)	460 564 (20.3)	0.317	124 626 (30.9)	124 627 (30.9)	<0.001
Mild mental illness[c]	82 640 (16.3)	238 225 (10.5)	0.170	59 537 (14.8)	59 537 (14.8)	<0.001
Hypertension	34 394 (6.8)	95 719 (4.2)	0.112	24 452 (6.1)	24 452 (6.1)	<0.001
Severe mental illness[c]	3663 (0.7)	9440 (0.4)	0.041	2545 (0.6)	2546 (0.6)	<0.001
Dyslipidemia	3503 (0.7)	9171 (0.4)	0.039	2429 (0.6)	2429 (0.6)	<0.001
Diabetes	2525 (0.5)	8209 (0.4)	0.021	1859 (0.5)	1859 (0.5)	<0.001
Severe maternal morbidity, No. (%)[d]						
0	460 242 (90.6)	2 108 338 (92.9)	0.074	368 125 (91.2)	368 126 (91.2)	<0.001
1	45 664 (9)	155 432 (6.9)		34 159 (8.5)	34 159 (8.5)	
≥2	1939 (0.4)	5504 (0.2)		1374 (0.3)	1374 (0.3)	
Hospital admissions in a year before pregnancy, No. (%)[e]						
0	415 693 (81.9)	1 913 083 (84.3)	0.064	332 318 (82.3)	332 319 (82.3)	<0.001
1	71 652 (14.1)	288 774 (12.7)		56 100 (13.9)	56 100 (13.9)	
≥2	20 500 (4.0)	67 417 (3.0)		15 240 (3.8)	15 240 (3.8)	
Outpatient contacts in a year before pregnancy, No. (%)[e]						
0	19 857 (3.9)	170 553 (7.5)	0.280	17 742 (4.4)	17 742 (4.4)	<0.001
1-4	88 452 (17.4)	575 002 (25.3)		76 253 (18.9)	76 253 (18.9)	
≥5	399 536 (78.7)	1 523 719 (67.2)		309 663 (76.7)	309 663 (76.7)	
Covariates not used in matching						
Mother						
Delivery type, No. (%)						
Vaginal	301 657 (59.4)	1 324 206 (58.4)	0.021	241 602 (59.9)	231 021 (57.2)	0.053
Cesarean	206 188 (40.6)	945 068 (41.7)		162 057 (40.2)	172 637 (42.8)	
Infant						
Sex, No. (%)						
Male	260 255 (51.3)	1 166 066 (51.4)	0.003	206 895 (51.3)	207 332 (51.4)	0.002
Female	247 590 (48.8)	1 103 208 (48.6)		196 763 (48.7)	196 327 (48.6)	

(continued)

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Table. Baseline Characteristics of Offspring in the Overlap-Weighted Cohort by Acid-Suppressive Medications Exposure (continued)

Characteristic	Unadjusted cohort (n = 2 777 119)[a]			Overlap-weighted cohort (n = 807 317)[a]		
	Exposed (n = 507 845)	Unexposed (n = 2 269 274)	SMD[b]	Exposed (n = 403 658)	Unexposed (n = 403 659)	SMD[b]
Season of birth, No. (%)						
Spring	128 049 (25.2)	592 918 (26.1)	0.028	102 089 (25.3)	104 445 (25.9)	0.028
Summer	127 401 (25.1)	554 062 (24.4)		101 105 (25.1)	99 153 (24.6)	
Autumn	126 050 (24.8)	557 609 (24.6)		99 672 (24.7)	101 136 (25.1)	
Winter	126 345 (24.9)	564 685 (24.9)		100 792 (25.0)	98 924 (24.5)	
At-risk newborn, No. (%)[f]						
Preterm birth	25 996 (5.1)	75 609 (3.3)	0.089	20 435 (5.1)	13 946 (3.5)	0.080
Low birth weight	19 828 (3.9)	61 401 (2.7)	0.067	15 623 (3.9)	11 205 (2.8)	0.061

[a] Each mother-child pair was defined as a unique birth; mothers with multiple deliveries could appear more than once.

[b] Standard mean difference (SMD) <0.1 indicates no significant imbalance.

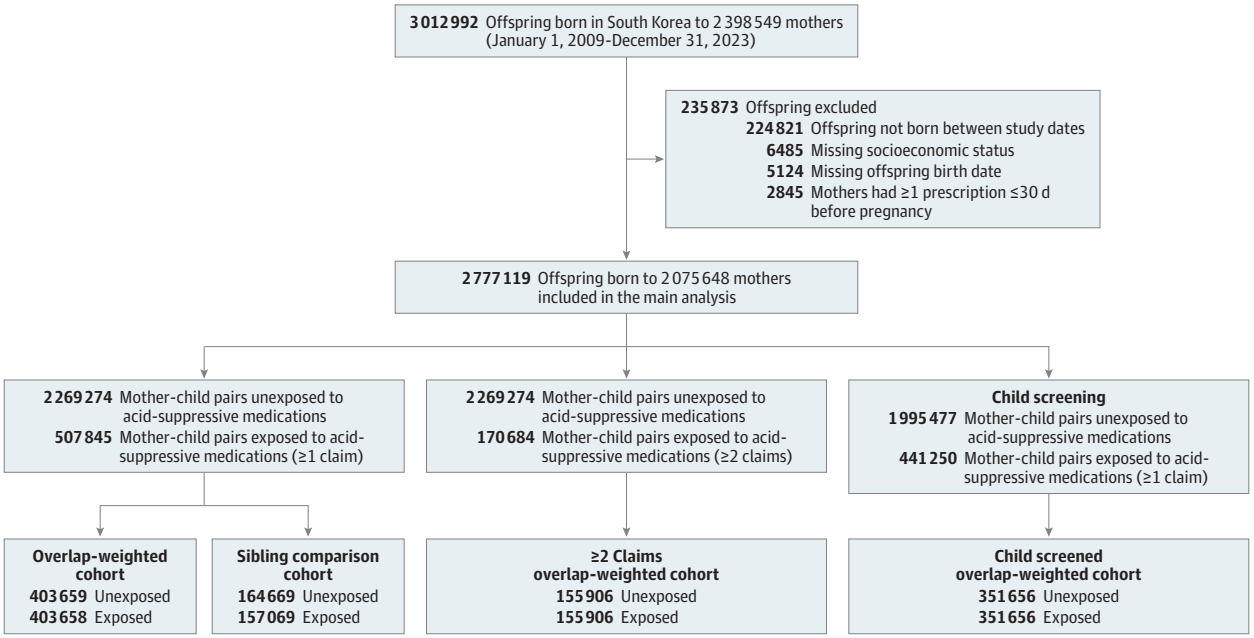
[c] Mild mental illness was defined as anxiety-related and stress-related disorders (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision [ICD-10]* codes F40-F48), alcohol or drug misuse (F10-F16, F18-F19), mood disorders without psychotic symptoms (F32-F34, F38-F39, excluding F32.3 and F33.3), eating disorders (F50), and personality disorders (F60-F63, F68-F69). Severe mental illness was defined as nonaffective or affective disorders with psychotic features, including nonaffective psychotic disorders (F20-F24, F28-F29), and affective psychotic disorders (F25, F30-F31, F32.3, F33.3).

[d] Severe maternal morbidity was defined according to the Centers for Disease Control and Prevention algorithm, encompassing 21 life-threatening maternal complications (eg, acute kidney failure, eclampsia, sepsis, cardiac arrest) identified by *ICD-10* diagnostic and procedural codes.

[e] Hospital admissions and outpatient contacts within a year before pregnancy, including mental or behavioral health–related encounters.

[f] Preterm birth was defined as delivery at 36 weeks or less of gestation, and low birth weight as 2499 g or less at birth.

Figure 1. Study Population for Evaluation of Prenatal Acid-Suppressive Medications and Offspring Neurodevelopment, Births in South Korea, 2010-2017



Each mother-child pair was defined as a unique birth; mothers with multiple deliveries could appear more than once.

Exposure

Maternal exposure to acid-suppressive medications was defined as having received 1 or more prescriptions for PPIs or H_2 receptor antagonists during pregnancy. The acid-suppressive medication exposure group was further classified into subgroups based on medication type (PPI or H_2 receptor antagonist) and timing of exposure (first, second, or third trimester).[19] The control group comprised women with no acid-suppressive medication prescriptions from 30 days prior to the start of pregnancy through delivery.

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Outcome

Outcomes were identified via *ICD-10* codes (eTable 2 in Supplement 1)[19] and included severe neuropsychiatric disorders (schizophrenia spectrum disorders and mood disorders with psychotic features), obsessive-compulsive disorders, attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorders (ASD), and intellectual disability. Onset was identified as the first postnatal diagnosis.

Covariates

Covariates included (1) maternal sociodemographic characteristics including maternal age at delivery, region of residence (rural or urban), household income level (by quartile), and parity (1 or ≥2 children); (2) maternal clinical characteristics, encompassing mental illness status (none, common, or severe); prepregnancy maternal medical conditions such as hypertension, diabetes, dyslipidemia, and gastrointestinal disorders; severity of mental illness (mild or severe); the number of severe maternal morbidities (0, 1, or ≥2); and health care utilization in the year before pregnancy, including hospital admissions (0, 1, or ≥2) and outpatient visits (0, 1-4, or ≥5); (3) delivery-related factors, including mode of delivery and season of birth; and (4) infant characteristics including sex, preterm birth, and low birth weight.

Statistical Analysis

A propensity score-based overlap-weighting algorithm was applied to adjust for measured baseline confounders, identified through hypothesized causal relationships using a directed acyclic graph, when comparing the acid-suppressive medication exposure group with the control group (eFigure in Supplement 2). Covariate balance between the 2 groups was assessed using standardized mean differences (SMDs), with an SMD <0.1 indicating no significant imbalance.

In the overlap-weighted cohort, Cox proportional hazards regression models were used to estimate adjusted hazard ratios (HRs) and 95% CIs for the incidence risk of neuropsychiatric disorders in offspring following prenatal exposure to acid-suppressive medications.[20] Subgroup analyses were conducted according to the type of acid-suppressive medication exposure (H_2 receptor antagonist only and PPI only) and exposed trimester (first, second, and third trimester). Differences in outcomes between drug classes and trimesters were assessed using a likelihood ratio test. To examine dose-response associations, we calculated the total patient-days receiving therapy by dividing the standard units of each drug by its defined daily dose. *P* values were adjusted for multiple testing using the Bonferroni-Holm method in the main analyses (α = .05). Analyses were conducted using SAS version 9.4 (SAS Institute Inc).

Sibling-Matched Analyses

In the sibling comparison analysis, stratified Cox proportional hazards regression models were used to account for within-family clustering, with each mother treated as a separate stratum (eTable 3 in Supplement 1). For mothers with 3 or more children, more than 1 child per mother could be included in the same group. Covariates with a SMD between the 2 groups exceeding 0.1 were adjusted for in the estimation of HRs.

Sensitivity Analysis

A sensitivity analysis in which exposure to the medication was redefined as at least 2 claims was conducted to reduce potential misclassification. Additionally, a child-screening sensitivity analysis was performed to evaluate whether the effects of maternal exposure persisted after excluding children with underlying health risks. In this analysis, children diagnosed with immune mechanism disorders, cystic fibrosis, chronic kidney disease, β-thalassemia or sickle cell disorders, malignancy, teratogenic or genetic syndromes, microdeletions, chromosomal abnormalities, and malformation syndromes were excluded (eTable 4 in Supplement 1).

Results

Study Cohort

Initially, 3 012 992 mother-child pairs were identified based on births from January 1, 2009, through December 31, 2023. After excluding mothers with missing data and those with a prescription for acid-suppressive medication within 30 days of pregnancy, 2 777 119 children born from January 1, 2010, through December 31, 2017, and 2 075 648 paired mothers were included; mean follow-up was 10.3 (SD, 2.3) years. Among them, 2 269 274 pairs were unexposed to acid-suppressive medications whereas 507 845 pairs were exposed. After applying propensity score-based overlap weighting, the cohort consisted of 807 317 pairs, with 403 658 pairs in the exposed group (Figure 1). The mean maternal age was 31.7 (SD, 3.8) years in the exposed group and 31.8 (SD, 1.8) years in the unexposed group. All baseline covariates had SMDs less than 0.1, indicating adequate balance (Table).

Prenatal Exposure and Risk of Neuropsychiatric Disorders

In overlap-weighted analyses, prenatal exposure to acid-suppressive medications was associated with a modestly increased incidence of any neuropsychiatric disorder with an adjusted HR of 1.13 (95% CI, 1.11-1.15) (**Figure 2**). ADHD was the most frequent outcome, with an adjusted HR of 1.14 (95% CI, 1.12-1.17). Elevated risks were also observed for severe neuropsychiatric disorder. The adjusted HR for neuropsychiatric disorder was 1.16 (95% CI, 1.10-1.21); for obsessive-compulsive disorder, 1.12 (95% CI, 1.03-1.21); for ASD, 1.07 (95% CI, 1.03-1.11); and for intellectual disability, 1.13 (95% CI, 1.09-1.18). *P* values for all outcomes were significant after adjustment for multiple testing (eTable 5 in Supplement 1). Sensitivity analyses using an exposure definition requiring at least 2 claims and a child-screened overlap-weighted cohort were consistent with the primary findings (eTables 6-9 in Supplement 1). The child-screened cohort showed increases across all neuropsychiatric disorders, whereas the 2 or more claims definitions showed increased risk for all except obsessive-compulsive disorder.

Subgroup Analyses

Subgroup analyses by medication type, cumulative defined daily dose, and trimester of exposure are presented in **Figure 3** (eTables 10 and 11 in Supplement 1). Prenatal exposure to either

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Figure 2. Association Between Prenatal Acid-Suppressive Medication Use and Children's Neuropsychiatric Disorder

Neuropsychiatric disorder	Unadjusted cohort						Overlap-weighted cohort					
	Exposed (507 845)		Unexposed (2 269 274)				Exposed (403 658)		Unexposed (403 659)			
	No. (%) of events	Incidence rate, per 1000 person-years	No. (%) of events	Incidence rate, per 1000 person-years	Adjusted risk difference[a] (95% CI), %	Adjusted hazard ratio[a] (95% CI)	No. (%) of events	Incidence rate, per 1000 person-years	No. (%) of events	Incidence rate, per 1000 person-years	Adjusted risk difference[b] (95% CI), %	Adjusted hazard ratio[b] (95% CI)
Any neuropsychiatric disorder	31 942 (6.29)	6.71	119 982 (5.29)	5.53	0.77 (0.71-0.84)	1.14 (1.12-1.15)	25 074 (6.21)	6.60	22 077 (5.47)	5.80	0.46 (0.47-0.65)	1.13 (1.11-1.15)
Attention-deficit/hyperactivity disorder	24 960 (4.91)	5.20	92 492 (4.09)	4.23	0.65 (0.59-0.70)	1.14 (1.13-1.16)	19 558 (4.85)	5.11	17 165 (4.25)	4.48	0.45 (0.37-0.52)	1.14 (1.12-1.17)
Autism spectrum disorder	7405 (1.46)	1.53	29 923 (1.32)	1.36	0.08 (0.05-0.11)	1.09 (1.06-1.12)	5855 (1.45)	1.52	5381 (1.33)	1.40	0.07 (0.03-0.11)	1.07 (1.03-1.11)
Intellectual disability	6468 (1.27)	1.34	24 095 (1.06)	1.10	0.16 (0.13-0.19)	1.15 (1.12-1.18)	5058 (1.25)	1.31	4393 (1.09)	1.14	0.12 (0.08-0.16)	1.13 (1.09-1.18)
Severe neuropsychiatric disorder[c]	4831 (0.95)	0.99	17 618 (0.78)	0.80	0.14 (0.11-0.16)	1.15 (1.12-1.19)	3786 (0.94)	0.98	3267 (0.81)	0.84	0.10 (0.07-0.13)	1.16 (1.10-1.21)
Obsessive-compulsive disorder	1559 (0.31)	0.32	6022 (0.27)	0.27	0.04 (0.03-0.06)	1.12 (1.06-1.18)	1227 (0.30)	0.32	1094 (0.27)	0.28	0.03 (0.01-0.05)	1.12 (1.03-1.21)

[a]The model was adjusted for maternal age (<20, 20-24, 25-29, 30-34, or ≥35 years), region of residence (urban or rural), household income (low, lower middle, upper middle, or high), parity (1 or ≥2 children), prepregnancy maternal medical conditions (gastrointestinal disorder, mild mental illness, hypertension, severe mental illness, dyslipidemia, and diabetes), severe maternal morbidity (0, 1, or ≥2), hospital admissions in a year before pregnancy (0, 1, or ≥2), outpatient contacts in a year before pregnancy (0, 1-4, or ≥5), delivery type (vaginal or cesarean delivery), infant's sex (male or female), season of birth (spring, summer, autumn, or winter), and at-risk newborn (preterm birth and low birth weight).

[b]The model was adjusted for delivery type (vaginal or cesarean delivery), infant's sex (male or female), season of birth (spring, summer, autumn, or winter), and at-risk newborn (preterm birth [yes or no] and low birth weight [yes or no]).

[c]Severe neuropsychiatric disorder was defined as nonaffective or affective disorders with psychotic features, including nonaffective psychotic disorders and affective psychotic disorders.

H_2 receptor antagonists only (adjusted HR, 1.12 [95% CI, 1.10-1.14]) or PPIs only (adjusted HR, 1.20 [95% CI, 1.13-1.26]) was associated with an increased risk of neuropsychiatric disorders. Differences between the PPI-only exposure and H_2 receptor antagonist-only exposure were not statistically significant. For H_2 receptor antagonist-only exposures, risks generally increased with higher cumulative defined daily dose. For PPI-only exposures, obsessive-compulsive disorder and intellectual disability were associated only with the highest tertile of cumulative defined daily dose, whereas ADHD and ASD were associated across all exposure levels.

By timing of the first exposure compared with unexposed offspring, initiation in the first trimester was consistently associated with higher risk of neuropsychiatric disorders (adjusted HR, 1.14 [95% CI, 1.12-1.17]). Specifically, elevated risks were reported for severe neuropsychiatric disorders (adjusted HR, 1.18 [95% CI, 1.13-1.24]), obsessive-compulsive disorders (adjusted HR, 1.14 [95% CI, 1.05-1.24]), ADHD (adjusted HR, 1.16 [95% CI, 1.14-1.19]), ASD (adjusted HR, 1.06 [95% CI, 1.02-1.10], and intellectual disability (adjusted HR, 1.13 [95% CI, 1.09-1.18]). In contrast, being first exposed during the second or third trimester was generally not associated with increased risks, although third trimester exposure was associated with ASD and intellectual disability. Additional analyses stratified by maternal age, region of residence, household income, prepregnancy maternal medical conditions, parity, and severe maternal morbidity score are presented in eTable 12 in Supplement 1.

Sibling-Matched Analysis

In the sibling comparison cohort comprising 151 737 discordant sibling pairs, 164 669 mother-child pairs were unexposed to acid-suppressive medications, whereas 157 069 pairs were exposed (eTable 13 in Supplement 1). In the sibling-matched prenatal exposure analyses, prenatal exposure to acid-suppressive medications was not significantly associated with any neuropsychiatric outcome (**Figure 4**). The incidence rate for any neuropsychiatric disorder was 5.72 and 5.77 per 1000 person-years in exposed and unexposed siblings, respectively, corresponding to an adjusted HR of 1.00 (95% CI, 0.97-1.03). No significant associations were found for individual disorders, including severe neuropsychiatric disorder with an adjusted HR of 1.00 (95% CI, 0.93-1.08); obsessive-compulsive disorder, 0.95 (95% CI, 0.82-1.10); ADHD, 0.98 (95% CI, 0.95-1.02); ASD, 0.98 (95% CI, 0.92-1.04); and intellectual disability, 1.02 (95% CI, 0.95-1.09; eTable 14 in Supplement 1). When stratified by maternal mental health history, no association was observed in either mothers with a history of mental health conditions (adjusted HR, 1.07 [95% CI, 0.79-1.46]) or

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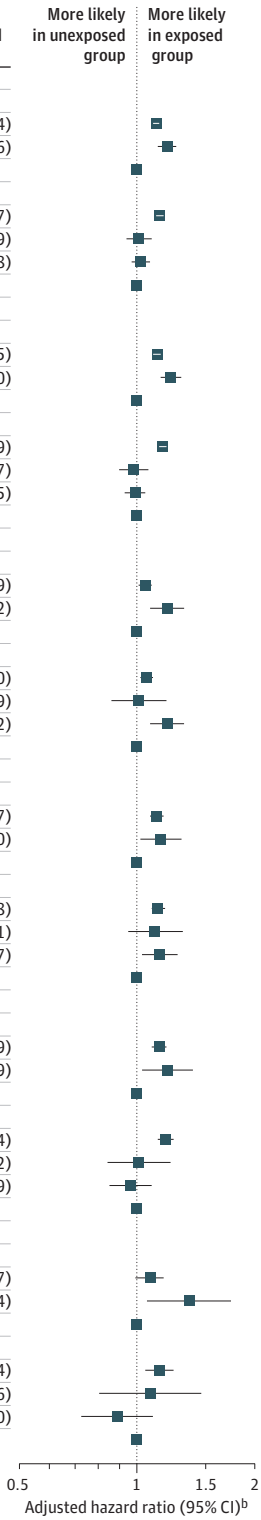
Figure 3. Association Between Prenatal Acid-Suppressive Medication Use and Children's Neuropsychiatric Disorder by Type and Trimester

Exposure	Unadjusted cohort (n = 2 777 119) No. of events/total (%)	Incidence rate, per 1000 person-years	Adjusted hazard ratio (95% CI)[a]	Overlap-weighted cohort (n = 807 317) No. of events/total (%)	Incidence rate, per 1000 person-years	Adjusted hazard ratio (95% CI)[b]
Any neuropsychiatric disorder						
Medication type						
H2RA only	28 520/456 679 (6.25)	6.59	1.12 (1.11-1.14)	22 475/364 186 (6.17)	6.49	1.12 (1.10-1.14)
PPI only	1825/29 378 (6.21)	7.39	1.28 (1.23-1.34)	1417/23 059 (6.14)	7.29	1.20 (1.13-1.26)
Unexposed	119 982/2 269 274 (5.29)	5.53	1 [Reference]	22 077/403 659 (5.47)	5.80	1 [Reference]
Trimester						
First	29 096/461 120 (6.31)	6.84	1.16 (1.14-1.17)	22 672/364 042 (6.23)	6.73	1.14 (1.12-1.17)
Second	832/13 568 (6.13)	5.65	1.00 (0.94-1.08)	697/11 448 (6.09)	5.61	1.01 (0.94-1.09)
Third	2014/33 067 (6.09)	5.60	1.01 (0.96-1.05)	1704/28 168 (6.05)	5.56	1.02 (0.97-1.08)
Unexposed	119 982/2 269 274 (5.29)	5.53	1 [Reference]	22 077/403 659 (5.47)	5.80	1 [Reference]
Attention-deficit/hyperactivity disorder						
Medication type						
H2RA only	22 319/456 679 (4.89)	5.11	1.12 (1.11-1.14)	17 555/364 186 (4.82)	5.03	1.13 (1.10-1.15)
PPI only	1402/29 378 (4.77)	5.63	1.31 (1.24-1.38)	1088/23 059 (4.72)	5.55	1.22 (1.15-1.30)
Unexposed	92 492/2 269 274 (4.08)	4.23	1 [Reference]	17 165/403 659 (4.25)	4.48	1 [Reference]
Trimester						
First	22 818/461 120 (4.95)	5.25	1.13 (1.07-1.20)	17 755/364 042 (4.88)	5.23	1.16 (1.14-1.19)
Second	627/13 568 (4.62)	4.16	1.15 (0.88-1.50)	524/11 448 (4.58)	4.17	0.98 (0.90-1.07)
Third	1515/33 067 (4.58)	4.10	0.94 (0.77-1.13)	1280/28 168 (4.54)	4.14	0.99 (0.93-1.05)
Unexposed	92 492/2 269 274 (4.08)	4.23	1 [Reference]	17 165/403 659 (4.25)	4.48	1 [Reference]
Autism spectrum disorder						
Medication type						
H2RA only	6458/456 679 (1.42)	1.47	1.06 (1.03-1.09)	5150/364 186 (1.41)	1.47	1.05 (1.01-1.09)
PPI only	517/29 378 (1.76)	2.07	1.34 (1.23-1.46)	403/23 059 (1.75)	2.05	1.20 (1.08-1.32)
Unexposed	29 923/2 269 274 (1.32)	1.36	1 [Reference]	5381/403 659 (1.33)	1.40	1 [Reference]
Trimester						
First	6728/461 120 (1.46)	1.57	1.16 (1.15-1.18)	5279/364 042 (1.45)	1.55	1.06 (1.02-1.10)
Second	176/13 568 (1.30)	1.18	0.97 (0.89-1.04)	149/11 448 (1.31)	1.18	1.01 (0.86-1.19)
Third	501/33 067 (1.52)	1.38	0.97 (0.92-1.02)	427/28 168 (1.51)	1.37	1.20 (1.08-1.32)
Unexposed	29 923/2 269 274 (1.32)	1.36	1 [Reference]	5381/403 659 (1.33)	1.40	1 [Reference]
Intellectual disability						
Medication type						
H2RA only	5770/456 679 (1.26)	1.31	1.14 (1.10-1.17)	4531/364 186 (1.24)	1.29	1.12 (1.08-1.17)
PPI only	366/29 378 (1.52)	1.46	1.24 (1.11-1.37)	282/23 059 (1.22)	1.43	1.15 (1.02-1.30)
Unexposed	24 095/2 269 274 (1.06)	1.10	1 [Reference]	4393/403 659 (1.09)	1.14	1 [Reference]
Trimester						
First	5833/461 120 (1.26)	1.35	1.09 (1.06-1.12)	4521/364 042 (1.24)	1.32	1.13 (1.09-1.18)
Second	185/13 568 (1.36)	1.24	0.94 (0.81-1.09)	155/11 448 (1.35)	1.22	1.11 (0.95-1.31)
Third	450/33 067 (1.36)	1.23	1.08 (0.99-1.18)	382/28 168 (1.35)	1.22	1.14 (1.03-1.27)
Unexposed	24 095/2 269 274 (1.06)	1.10	1 [Reference]	4393/403 659 (1.09)	1.14	1 [Reference]
Severe neuropsychiatric disorder[c]						
Medication type						
H2RA only	4353/456 679 (0.95)	0.99	1.14 (1.10-1.18)	3423/364 186 (0.94)	0.97	1.14 (1.09-1.19)
PPI only	238/29 378 (0.81)	0.95	1.25 (1.10-1.42)	186/23 059 (0.81)	0.94	1.20 (1.03-1.39)
Unexposed	17 618/2 269 274 (0.78)	0.80	1 [Reference]	3267/403 659 (0.81)	0.84	1 [Reference]
Trimester						
First	4359/461 120 (0.95)	1.02	1.16 (1.14-1.17)	3388/364 042 (0.93)	0.99	1.18 (1.13-1.24)
Second	144/13 568 (1.06)	0.98	1.00 (0.94-1.08)	121/11 448 (1.05)	0.95	1.01 (0.84-1.22)
Third	328/33 067 (0.99)	0.91	1.01 (0.96-1.05)	277/28 168 (0.98)	0.89	0.96 (0.85-1.09)
Unexposed	17 618/2 269 274 (0.78)	0.80	1 [Reference]	3267/403 659 (0.81)	0.84	1 [Reference]
Obsessive-compulsive disorder						
Medication type						
H2RA only	1381/456 679 (0.30)	0.31	1.08 (1.02-1.15)	1091/364 186 (0.30)	0.31	1.08 (0.99-1.17)
PPI only	85/29 378 (0.29)	0.34	1.38 (1.12-1.72)	67/23 059 (0.29)	0.34	1.36 (1.06-1.74)
Unexposed	6022/2 269 274 (0.27)	0.27	1 [Reference]	1094/403 659 (0.27)	0.28	1 [Reference]
Trimester						
First	1398/461 120 (0.30)	0.32	1.17 (1.13-1.21)	1091/364 042 (0.3)	0.32	1.14 (1.05-1.24)
Second	54/13 568 (0.40)	0.36	1.04 (0.88-1.23)	45/11 448 (0.39)	0.35	1.08 (0.80-1.46)
Third	107/33 067 (0.32)	0.29	0.99 (0.89-1.11)	90/28 168 (0.32)	0.29	0.89 (0.72-1.10)
Unexposed	6022/2 269 274 (0.27)	0.27	1 [Reference]	1094/403 659 (0.27)	0.28	1 [Reference]



[a]For the adjusted conditions, see footnote *a* in the Figure 2 legend.

[b]For the adjusted conditions, see footnote *b* in the Figure 2 legend.

[c]For the definition of severe neuropsychiatric disorders, see footnote *c* in the Figure 2 legend.

H2RA indicates histamine 2 receptor antagonists; PPI, proton pump inhibitor.

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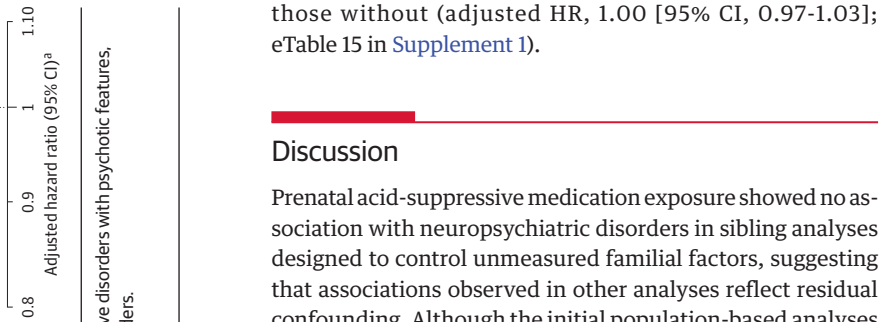
Figure 4. Neuropsychiatric Risk in Children After Prenatal Acid-Suppressive Medication Exposure in a Sibling Comparison Cohort

Neuropsychiatric disorder	Exposed cohort (157 069)		Unexposed cohort (164 669)		Crude		Adjusted[a]	
	No. (%) of events	Incidence rate, per 1000 person-years	No. (%) of events	Incidence rate, per 1000 person-years	Risk difference (95% CI)	Hazard ratio (95% CI)	Risk difference (95% CI)	Hazard ratio (95% CI)
Any neuropsychiatric disorder	8274 (5.27)	5.72	9158 (5.56)	5.77	−0.29 (−0.45 to 0.14)	1.02 (0.99 to 1.05)	−0.01 (−0.16 to 0.14)	1.00 (0.97 to 1.03)
Attention-deficit/hyperactivity disorder	6498 (4.14)	4.46	7388 (4.49)	4.62	−0.35 (−0.49 to −0.21)	1.00 (0.97 to 1.03)	−0.12 (−0.25 to 0.02)	0.98 (0.95 to 1.02)
Autism spectrum disorder	1766 (1.12)	1.20	1910 (1.16)	1.19	−0.04 (−0.11 to 0.04)	0.99 (0.93 to 1.06)	0.00 (−0.07 to 0.08)	0.98 (0.92 to 1.04)
Intellectual disability	1675 (1.07)	1.14	1747 (1.06)	1.08	0.01 (−0.07 to 0.08)	1.06 (0.99 to 1.13)	0.02 (−0.05 to 0.09)	1.02 (0.95 to 1.09)
Severe neuropsychiatric disorder[b]	1214 (0.77)	0.82	1402 (0.85)	0.87	−0.08 (−0.14 to −0.02)	0.97 (0.89 to 1.04)	0.01 (−0.05 to 0.07)	1.00 (0.93 to 1.08)
Obsessive-compulsive disorder	341 (0.22)	0.23	439 (0.27)	0.27	−0.05 (−0.08 to −0.02)	0.94 (0.82 to 1.08)	−0.08 (−0.25 to 0.09)	0.95 (0.82 to 1.10)



[a]The model was adjusted for maternal age (<20, 20-24, 25-29, 30-34, or ≥35 years), parity (1 or ≥2 children), prepregnancy maternal medical conditions (gastrointestinal disorder, mild mental illness, and severe mental illness), and outpatient contacts in a year before pregnancy (0, 1-4, ≥5).

[b]Severe neuropsychiatric disorder was defined as nonaffective or affective disorders with psychotic features, including nonaffective psychotic disorders and affective psychotic disorders.

those without (adjusted HR, 1.00 [95% CI, 0.97-1.03]; eTable 15 in Supplement 1).

Discussion

Prenatal acid-suppressive medication exposure showed no association with neuropsychiatric disorders in sibling analyses designed to control unmeasured familial factors, suggesting that associations observed in other analyses reflect residual confounding. Although the initial population-based analyses of more than 2.7 million mother-child pairs suggested small increases in most neuropsychiatric disorders, the absolute excess risks were extremely low. PPI and H_2 receptor antagonist exposures showed similar risk profiles without statistically significant differences between medication classes.

This study's findings that prenatal acid-suppressive medication exposure showed no significant association with children's neurodevelopment in real-world human populations stand in contrast to studies in animal models that have shown that alterations in the maternal gut microbiome may impair neurodevelopment and axon growth.[11,21-25] The current study's results also differ from studies of postnatal acid-suppressive medications, which have reported elevated neuropsychiatric risks.[12,26]

This study's sibling analysis revealed no significant associations between either PPIs or H_2 receptor antagonists and child neuropsychiatric disorders, suggesting these relationships may reflect confounding rather than causation. To our knowledge, no direct causal pathway linking prenatal acid-suppressive medications to offspring neuropsychiatric disorders has been established in humans.

Limitations

This study has several limitations. First, exposure misclassification may have occurred because prescription records do not guarantee medication adherence. Although a sensitivity analysis defining exposure as more than 2 prescriptions was conducted to approximate consistent use, pharmacy-dispensing data were unavailable and medication intake could not be directly confirmed. Second, as in any observational study, residual confounding may have persisted despite rigorous adjustment. The full-cohort model is susceptible to unmeasured confounders. Although sibling analyses may mitigate shared familial factors, sibling analyses are vulnerable to time-varying confounders. Furthermore, although multiple covariates were included, data were limited on prepregnancy body mass index and maternal lifestyle, which are potential risk factors for gastroesophageal reflux disease and subsequent acid-suppressive medication use. To address this, several prepregnancy maternal conditions including pregestational diabetes and dyslipidemia were incorporated. Third, genetic factors were not incorporated. Certain genetic variants are associated with increased neuropsychiatric disorder risk,[27] and differential expression of placental transporters, resulting in varying levels of fetal acid-suppressive medication exposure.[28] Fourth, underestimation from unrecorded over-the-counter (OTC) use cannot be completely excluded.

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However, in South Korea, PPIs require prescription, and OTC access to H_2 receptor antagonists are highly restricted to low-dose famotidine and cimetidine, with higher doses and other H_2 receptor antagonists requiring a prescription.[29] Furthermore, South Korea's universal single-payer health insurance system reduces financial barriers, ensures broad health care access, and systematically captures prescriptions,[30] resulting in high antenatal care engagement further supporting prescription-based management.[31] Fifth, the study cohort included only live births, precluding assessment of associations with pregnancy loss or stillbirth. However, previous studies report no significant associations between PPI use and stillbirth or neonatal death, suggesting minimal selection bias.[32] Sixth, trimester-specific and stratified analyses should be interpreted cautiously due to potential residual confounding from distinct covariate patterns in exposure subgroups that cannot be replicated in the unexposed population, and the absence of prespecified hypotheses for certain subgroups. Seventh, because the current findings are limited to Asian populations, differences in health care systems, prescribing patterns, and genetic backgrounds may limit generalizability, necessitating validation in other populations.

Conclusions

In this large nationwide birth cohort study, exposure to acid-suppressive medication during pregnancy was not associated with children's risk of ADHD, severe neuropsychiatric disorder, obsessive-compulsive disorder, intellectual disability, or ASD in sibling-control analysis. Associations observed in other models may reflect confounding by shared familial factors.

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Author Affiliations: Department of Medicine, Kyung Hee University College of Medicine, Seoul, South Korea (Hong, S. Lee, T. H. Kim, Yon); Center for Digital Health, Medical Science Research Institute, Kyung Hee University Medical Center, Kyung Hee University College of Medicine, Seoul, South Korea (Hong, S. Lee, H. Kim, H. Jo, K. Lee, Y. Jo, T. H. Kim, Park, H. Lee, Yon); Department of Precision Medicine, Kyung Hee University School of Medicine, Seoul, South Korea (H. Kim, Y. Jo, Park, Yon); Department of Regulatory Science, Kyung Hee University, Seoul, South Korea (H. Jo, K. Lee, Yon); Department of Biomedical Engineering, Kyung Hee University, Yongin, South Korea (J. Lee, H. Lee, Yon); Department of Electronics and Information Convergence Engineering, Kyung Hee University, Yongin, South Korea (J. Lee, H. Lee); Department of Neurology, Kyung Hee University Medical Center, Kyung Hee University College of Medicine, Seoul, Korea (Woo); Department of Pediatrics, Kyung Hee University Medical Center, Kyung Hee University College of Medicine, Seoul, South Korea (Yon).

Author Contributions: Dr Yon had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Ms Hong and Dr S. Lee contributed equally. Drs H. Lee and Yon jointly supervised this work.
Concept and design: Hong, S. Lee, H. Kim, H. Jo, K. Lee, Y. Jo, T. Kim, Park, Woo, H. Lee, Yon.
Acquisition, analysis, or interpretation of data: J. Lee, Woo, Yon.
Drafting of the manuscript: S. Lee, Yon.
Critical review of the manuscript for important intellectual content: Hong, H. Kim, H. Jo, K. Lee, Y. Jo, T. Kim, Park, J. Lee, Woo, H. Lee, Yon.
Statistical analysis: Park, Yon.
Obtained funding: Woo, Yon.
Administrative, technical, or material support: Hong, H. Kim, H. Jo, K. Lee, Y. Jo, T. Kim, J. Lee, Woo, H. Lee, Yon.
Supervision: S. Lee, Woo, Yon.

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